10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 21 — Space and Environment

Solved Exercise | New Book (PECTAA 2026)

Exercise 21 (Solved)

This exercise covers the Sun as a fusion reactor, the planets and their characteristics, global warming and the greenhouse effect, cyclones and the Coriolis effect, background radiation and the safe handling of radioactive materials.

A. Multiple Choice Questions

21.1 What process generates energy in the Sun?

(a) Nuclear fission
(b) Nuclear fusion
(c) Radioactive decay
(d) Chemical reactions

Answer: (b) Nuclear fusion — in the Sun's core hydrogen nuclei fuse into helium nuclei, and the mass defect appears as energy according to $E = mc^2$.

21.2 The factor which primarily determines a planet's surface temperature:

(a) its density
(b) its orbital distance from the Sun
(c) its gravitational field strength
(d) the number of moons it has

Answer: (b) its orbital distance from the Sun — the intensity of solar radiation falls as $1/r^2$, so the farther a planet is from the Sun, the less energy it receives and the colder its surface.

21.3 What is the primary reason cyclones form over warm ocean waters?

(a) The presence of strong winds
(b) The high atmospheric pressure
(c) The rising of warm, moist air creating low pressure
(d) The gravitational pull of the moon

Answer: (c) The rising of warm, moist air creating low pressure — warm sea water evaporates and the moist air rises, leaving a low-pressure centre; the inflowing air and the latent heat released on condensation build up the storm.

21.4 Which of the following gas contributes the most to global warming?

(a) Oxygen
(b) Carbon dioxide
(c) Nitrogen
(d) Argon

Answer: (b) Carbon dioxide — carbon dioxide, released in huge quantities by burning fossil fuels, is the main greenhouse gas added by human activity.

21.5 How does the Coriolis effect influence cyclones?

(a) It causes cyclones to spin in opposite directions in different hemispheres
(b) It strengthens cyclones over land
(c) It prevents warm air from rising
(d) It stops the formation of low-pressure areas

Answer: (a) It causes cyclones to spin in opposite directions in different hemispheres — because of the Earth's rotation, moving air is deflected to the right

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in the northern hemisphere and to the left in the southern, so cyclones spin anticlockwise in the north and clockwise in the south.

21.6 A major cause of rising sea levels is:

(a) increased volcanic activity
(b) the Earth moving closer to the Sun
(c) melting glaciers and ice sheets
(d) decreased solar radiation

Answer: (c) melting glaciers and ice sheets — global warming melts land ice and also makes sea water expand as it warms, and both raise the sea level.

21.7 A significant source of background radiation is:

(a) oxygen in the air
(b) cosmic rays from space
(c) sound waves in the atmosphere
(d) light from the Sun

Answer: (b) cosmic rays from space — cosmic rays from space and natural radioactive isotopes in rocks, soil and our own bodies are the main natural sources of the background radiation around us.

B. Short Answer Questions

21.1 What are the two main elements that make up the Sun?

The Sun is made up mainly of hydrogen and helium — about 71 % hydrogen and about 27 % helium by mass, with only about 2 % of heavier elements such as oxygen, carbon, neon and iron.

The hydrogen in the core is the fuel of the Sun: at a temperature of about 15 million kelvin four hydrogen nuclei fuse to form one helium nucleus, releasing energy. The helium so formed is the ash of this fusion. The Sun is a giant ball of plasma held together by its own gravity, and the balance between the outward pressure of radiation and the inward pull of gravity keeps it stable.

21.2 What factors are used to compare planets in terms of their characteristics?

Planets are compared by their size (diameter), mass and density; their distance from the Sun and their orbital period (year) and rotational period (day); their surface temperature and the energy they receive from the Sun; the composition of their atmosphere and its pressure; the number of satellites (moons) and rings; their gravitational field strength; the presence of water and of a magnetic field; and their surface features — craters, volcanoes, oceans or ice.

On the basis of these, the planets fall into two clear groups. The inner rocky (terrestrial) planets — Mercury, Venus, Earth and Mars — are small, dense, have solid surfaces and few or no moons. The outer giant planets — Jupiter, Saturn, Uranus and Neptune — are much larger, have low density, thick atmospheres of hydrogen and helium, and many moons and rings. Such comparison helps us to classify the planets and to understand their evolution.

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21.3 Explain how global warming increases the intensity of hurricanes.

A hurricane (tropical cyclone) takes its energy from warm ocean water. Global warming raises the temperature of the sea surface, and water above about 26.5 °C provides much more evaporation. More evaporation means more warm, moist air rising, and when this vapour condenses at a height it releases a large amount of latent heat. This heat warms the air further, makes it rise faster and lowers the pressure at the centre more, so more air rushes in — the circulation becomes stronger.

The result is a hurricane with faster winds, heavier rainfall and a larger storm surge. Moreover, warmer air can hold more moisture — about 7 % more for every 1 °C of warming — so the total rainfall of a storm increases and flooding becomes more likely. Because the sea stays warm for a longer part of the year, storms can also form over a longer season and can reach higher latitudes.

So global warming does not necessarily produce more storms, but it makes the strong ones stronger and wetter, and the damage they cause much greater.

21.4 What is background radiation, and what are its natural sources?

Background radiation is the low level of ionizing radiation that is always present in our environment from natural and artificial sources. Every person receives a small dose of it, and it is unavoidable.

Natural sources: (i) cosmic rays from space, which increase with altitude — people in the mountains or in aircraft receive more; (ii) radioactive isotopes in the rocks and soil, mainly potassium-40, uranium-238, thorium-232 and their decay products such as radon gas; (iii) radon gas seeping out of the ground into buildings, which is the largest single natural source; (iv) radioactive carbon-14 and potassium-40 present in our own bodies and in plants; and (v) small amounts from food, water and the air.

Artificial sources: medical X-rays and scans, radioactive tracers used in diagnosis, radioactive waste and fallout from past nuclear tests, and small amounts from nuclear power stations and from some smoke detectors and luminous paints. About 80 % of the average dose comes from natural sources.

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21.5 Name three safety measures used to handle radioactive materials safely.

1. Shielding: radioactive sources are kept in thick-walled lead containers, and gamma sources are handled behind lead or concrete shields, because lead absorbs the radiation and does not let it reach the worker.

2. Distance and remote handling: because the dose falls as the inverse square of the distance, sources are handled with long tongs or in a remote handling cell, and workers stay as far away as possible for as short a time as possible.

3. Protective clothing and monitoring: workers wear lead aprons, gloves and film badges or dosimeters that record their exposure; the material is stored in a labelled, locked room; contaminated waste is disposed of according to the rules; and eating, drinking or smoking is forbidden in the laboratory. (These three rules — shielding, distance and time — are the standard precautions.)

21.6 Why do cyclones weaken when they move over land?

A cyclone draws its energy from the evaporation of warm sea water. When it moves over land, this supply of moisture and latent heat is cut off, so the storm loses its source of energy.

There is also much more friction over land. The forests, hills and buildings slow down the surface winds and disrupt the tight circulation round the low-pressure centre, so the pressure gradient weakens. Without the warm water beneath it, the rising air currents weaken, the eye fills in and the storm decays into an ordinary low-pressure system.

That is why coastal areas suffer the full strength of a cyclone while inland areas receive only heavy rain and strong winds, and why a cyclone that stays over the sea for a long time becomes stronger and stronger.

21.7 How does the Coriolis Effect influence the movement of cyclones?

The Earth rotates from west to east, so a point near the equator moves faster than a point near the poles. Air that moves over the Earth's surface therefore appears to be deflected: to the right in the northern hemisphere and to the left in the southern hemisphere. This apparent deflection is called the Coriolis effect, and it is zero at the equator and greatest at the poles.

In a cyclone, air flows inward towards the low-pressure centre. Because of the Coriolis effect, this inward flow does not go straight to the centre but is turned sideways, so the winds spiral round the centre: anticlockwise in the northern hemisphere and clockwise in the southern. It is this effect that gives the cyclone its circular shape and its spinning motion.

Without the Coriolis effect the air would simply rush straight into the low-pressure region and no circulating storm could form. For the same reason cyclones do not form within about 5° of the equator, where the effect is too weak.

21.8 What is the primary cause of rising sea levels due to global warming?

There are two main causes, both due to global warming. The first and larger one is the melting of land ice — glaciers in the mountains and the huge ice sheets of Greenland and Antarctica. The melt water runs into the sea and adds to its volume. (Melting of floating sea ice does not raise the level, because it is already displacing its own weight of water.)

The second cause is thermal expansion. As the ocean water warms it expands slightly, and because the oceans are so deep and wide this expansion adds up to a significant rise. About half of the observed rise in the past century is due to this effect, and half to melting ice.

The consequences are serious for low-lying coasts: flooding, erosion, the intrusion of salt water into the rivers and wells, the loss of land in countries such as Bangladesh and the Maldives, and the need to build sea walls and embankments. That is why reducing greenhouse gas emissions is so important.

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C. Constructed Response Questions

21.1 Why do gas giants like Jupiter, despite their larger size, have lower densities than rocky planets such as Earth and Mars? What does this tell us about their internal structure?

Density is mass divided by volume. The rocky planets such as Earth and Mars are made mainly of silicates and iron, which are heavy materials, packed into a small volume; their densities are therefore high — about 3.9 g cm^{-3} for Mars and 5.5 g cm^{-3} for Earth. Jupiter and Saturn are made mainly of hydrogen and helium, the lightest elements, with only a comparatively small core of rock and ice. Even though Jupiter is about 11 times wider than the Earth and has 318 times its mass, its volume is about 1,300 times larger, so its average density is only about 1.3 g cm^{-3} — less than that of water.

What this tells us about internal structure: the gas giants have no solid surface to stand on. They consist of a thick outer envelope of hydrogen and helium gas, which gradually becomes liquid and then metallic hydrogen under the enormous pressure inside, with a small rocky or icy core at the centre. Their heat comes partly from the Sun and partly from their own slow contraction. In contrast, the rocky planets have a thin atmosphere over a solid crust, a mantle of dense rock and a metallic core of iron and nickel.

So the low density of the giants is direct evidence of their composition and of the enormous compression in their interiors, and it is the main reason why they are called gas giants.

21.2 If two planets receive the same amount of sunlight but have different atmospheric compositions, why might their surface temperatures still differ significantly?

The temperature of a planet's surface depends not only on how much sunlight it receives but also on how much of that energy is reflected back and how much of the outgoing radiation is trapped. Both are decided by the atmosphere.

Albedo (reflection). Clouds, ice and light-coloured dust reflect a part of the incident sunlight straight back into space. Venus reflects about 70 % of the sunlight falling on it, although it is nearer the Sun than the Earth; yet its surface is the hottest of all the planets, because of its atmosphere. A planet with bright clouds stays cool; a planet with a clear, dark atmosphere absorbs more and stays warm.

Greenhouse effect (trapping). An atmosphere rich in carbon dioxide, water vapour or methane is transparent to the incoming short-wave sunlight but absorbs the long-wave infrared radiated by the surface, and sends part of it back. Venus has an atmosphere of 96 % carbon dioxide at 90 times the Earth's pressure, and the resulting runaway greenhouse effect keeps its surface at about 465 °C, hotter than Mercury, which is much closer to the Sun. Mars has only a thin carbon dioxide atmosphere (about 1 % of the Earth's pressure), so its greenhouse effect is weak and its surface is very cold despite being the next planet outward.

So two planets at the same distance can differ greatly in temperature because one has a thick insulating atmosphere and the other a thin one. The Earth's atmosphere,

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with its moderate carbon dioxide and water vapour, keeps the surface about 33 °C warmer than it would otherwise be — a good example of the balance that makes a planet habitable.

21.3 How does the coriolis effect play a role in creating the spiralling motion of cyclones, and what might happen to cyclone formation if the Earth stopped rotating?

In a cyclone the air rushes inward towards a low-pressure centre. If the Earth did not rotate, this air would move in a straight line, directly towards the centre, and there would be no spiralling: the wind would blow radially inward and the air would rise at the centre.

Because the Earth rotates, a parcel of air is deflected — to the right in the northern hemisphere and to the left in the southern — and the deflection is greater the faster the air moves and the farther it is from the equator. The inflow is therefore continuously turned sideways, and the combination of the inward pressure-gradient force and the sideways Coriolis force makes the air follow a curved path. The result is the familiar spiral: anticlockwise in the northern hemisphere and clockwise in the southern, with the strongest winds near the centre and calm air in the eye.

If the Earth stopped rotating, the Coriolis effect would vanish. Air would flow straight into the low-pressure region and rise there; heavy rain could still occur, but no rotating storm could form — there would be no cyclones, hurricanes or tornadoes as we know them, and no trade winds or prevailing westerlies either. Instead, the winds would be simply a direct flow from cold regions to hot regions. The weather patterns of the whole planet would be completely different.

21.4 Compare the ways radiation, conduction, and convection each contribute to geothermal activity. Why is convection considered the primary driver of tectonic plate motion?

Geothermal activity means the transfer of heat from the Earth's hot interior to the surface. All three methods of heat transfer play a part, but in very different ways.

Radiation. Heat from the decay of radioactive isotopes (uranium, thorium, potassium) in the crust and mantle provides the energy source itself, and at the surface a small part of the heat escapes by radiation into space. Radiation is important as a source of heat rather than as a means of moving it through the solid rock, because rock is opaque and radiation is absorbed within a very short distance.

Conduction. Heat is conducted slowly through the solid rock of the crust and the lithosphere; the temperature gradient in the crust is about 25 °C per km. Conduction is a slow process, but it is significant in thin crustal layers, in the cooling of the lithosphere and in volcanic and hydrothermal areas where heat reaches the surface by conduction through the rock.

Convection. In the mantle, the rock is hot enough (1,000–1,300 °C) to flow slowly like a very stiff liquid over long times. The hot rock near the core rises, spreads sideways at the top, cools and sinks again, forming huge convection cells. This is by far the most effective way of carrying heat through the 2,900 km thick mantle. These convection currents drag the plates of the lithosphere along, push them apart

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at mid-ocean ridges and pull them down at subduction zones.

That is why convection is considered the primary driver of plate motion: it is the only mechanism that can move the enormous mass of the plates, it explains the pattern of earthquakes, volcanoes and mountain building at plate boundaries, and it matches the measured rate of plate movement of a few centimetres per year. So radiation supplies the heat, conduction moves a little of it through the crust, and convection does the main work of transferring heat and of moving the continents.

21.5 Explain why background radiation is always present in our environment and how cosmic and terrestrial sources balance each other despite fluctuations in human activity.

Background radiation is always present because its sources are permanent features of our world. The Earth's rocks and soil contain long-lived radioactive isotopes — uranium-238, thorium-232, potassium-40 and their daughter products, especially radon gas — which have half-lives of hundreds of millions to billions of years and therefore keep radiating. Cosmic rays from the Sun and from outer space continuously bombard the atmosphere, and our own bodies contain carbon-14 and potassium-40. These natural sources cannot be switched off, so a certain dose is received by everyone, everywhere, all the time.

The cosmic and terrestrial contributions are not equal everywhere, but they balance in the sense that the total background stays roughly constant. High up in the mountains or in an aircraft the air above is thinner, so the cosmic-ray dose increases; but the mass of rock and soil below is smaller, so the terrestrial dose from the ground decreases. In a deep valley or a cellar the reverse is true: little cosmic radiation reaches it, but the radon and gamma dose from the surrounding rock is higher. So a rise in one contribution is usually matched by a fall in the other, and the average total dose stays nearly the same.

Human activity adds a comparatively small amount — medical X-rays, nuclear medicine, fallout from past tests and discharges from nuclear plants — and the level from these has changed over the years and is carefully monitored. Man-made sources do not destroy the natural balance; they simply add a small amount to a much larger natural background, which is why background radiation has to be measured and subtracted in every radiation experiment.

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D. Comprehensive Questions

21.1 Describe the process of nuclear fusion in the Sun and explain how it generates energy.

Conditions in the core. The Sun's core has a temperature of about 15 million kelvin and a pressure of billions of atmospheres, and it is made almost entirely of hydrogen. At such a temperature the hydrogen atoms are stripped of their electrons and exist as a plasma of bare protons and electrons, moving at very high speeds.

The proton–proton chain. Normally the electrostatic repulsion between two protons would keep them apart, but at this temperature they collide so violently that they come close enough for the strong nuclear force to act. The sequence is: two protons

fuse to form a deuteron, releasing a positron and a neutrino; the deuteron absorbs another proton to form helium-3, releasing a gamma photon; and two helium-3 nuclei fuse to give helium-4 with the release of two protons. The net result of the chain is:

$$4\ {}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\ e^{+} + 2\ \gamma + 2\ \nu + 26.7\ \mathrm{MeV}$$

Where the energy comes from. The mass of the helium-4 nucleus (4.0026 u) is slightly less than the mass of four protons (4 × 1.0073 = 4.0292 u). The difference, about 0.0266 u per helium nucleus, is the mass defect. It does not disappear: by Einstein's equation $E = mc^2$ it appears as energy — about 26.7 MeV for each helium nucleus formed, which is millions of times the energy released in a chemical reaction.

How the energy reaches us. About 4 million tonnes of mass is converted into energy every second. The energy is first carried outward from the core by gamma rays, then by convection through the outer layers, and finally leaves the visible surface (the photosphere) as light and heat. It takes roughly 100,000 years for the energy to work its way out, but then travels the 150 million kilometres to the Earth in only 8 minutes 20 seconds. The outward radiation pressure balances the inward pull of gravity, so the Sun stays stable. This fusion has kept the Sun shining for about 4.6 billion years and will keep it shining for billions more.

21.2 How does a planet's orbital distance affect its surface temperature and the amount of solar energy it receives?

The Sun radiates energy equally in all directions, so the energy falling on unit area per second (the solar constant, or irradiance) decreases with the square of the distance: $I = P / (4\pi r^2)$, where P is the Sun's total power output. A planet at twice the Earth's distance therefore receives only one quarter of the energy per square metre.

This directly decides the equilibrium temperature. In the steady state the energy a planet absorbs must equal the energy it radiates away as infrared radiation. Using the Stefan–Boltzmann law, the temperature T of the planet is given by $T^4 \propto (1 - a) / r^2$, where a is the albedo. So the surface temperature falls as the distance increases: Mercury, at 0.39 AU, has a day-side temperature of about 430 °C; the Earth, at 1 AU, averages about 15 °C; Mars, at 1.52 AU, averages about −60 °C; and Neptune, at 30 AU, is about −215 °C. This is why the inner planets are rocky and hot while the outer planets are frozen.

Distance also decides the kind of material that can exist on a planet. Close to the Sun only rocks and metals could condense, because the ices would evaporate; farther out, water, ammonia and methane ices could survive, which is why the giants are rich in these substances and have thick atmospheres. The length of the year also increases with distance, from 88 days for Mercury to 165 years for Neptune. Distance alone does not fix the temperature, of course — the atmosphere also matters, as Venus with its runaway greenhouse effect at 465 °C shows — but it is the primary factor.

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21.3 Explain the role of convection in the formation and strengthening of cyclones.

A cyclone begins over a warm tropical ocean where the surface temperature is above about 26.5 °C. The water evaporates rapidly, so the air just above the sea becomes warm and very humid. Being warm, this air is less dense than the air above it, and it rises — this is convection. As it rises, the water vapour condenses into clouds and droplets, and condensation releases latent heat. This heat warms the air column further, so it rises even faster.

The rising air leaves a region of low pressure at the surface. Air from the surrounding area flows in to fill the gap, is warmed by the warm sea and rises in turn, so a continuous chain of convection currents is set up. The Coriolis effect turns this inflow into a spiral, and the storm begins to rotate. Because the air is continuously fed with moisture from the warm sea, the convection does not die out; instead it strengthens, the pressure at the centre falls further and the winds blow faster.

The structure of a mature cyclone shows convection clearly: in the eye wall the moist air rises very rapidly, forms tall cumulonimbus clouds and produces torrential rain; at the top of the troposphere the air spreads outward and sinks in the calm, clear eye; and at the surface the winds spiral inward. The energy of the whole storm comes from the latent heat released by condensation, which is why a cyclone weakens as soon as it leaves the warm ocean and moves over land, where this supply is cut off. So convection is both the trigger and the engine of the cyclone.

21.4 What are the major sources of background radiation, and how does human activities contribute to it?

Natural sources make up about 80 % of the dose that an average person receives.

1. Radon gas — the largest single source (about half of the natural dose). Radon-222 is a radioactive gas produced in the decay of uranium in rocks and soil; it seeps into houses through floors and walls and is inhaled, where its alpha-emitting daughters irradiate the lungs.

2. Terrestrial radiation — gamma rays from potassium-40, uranium-238, thorium-232 and their decay products in soil, rocks and building materials such as granite and brick.

3. Cosmic rays — high-energy particles from the Sun and from supernovae, which produce secondary radiation in the atmosphere. The dose increases with altitude (about 0.1 mSv per year extra for every 1,000 m) and is much higher for aircrew and astronauts.

4. Internal radiation — from potassium-40 and carbon-14 naturally present in our own bodies and in our food.

Human activities add the remaining 20 %, mainly: medical uses — X-rays, CT scans, nuclear medicine and radiotherapy — which form the largest artificial source; fallout and waste from past nuclear weapon tests and accidents such as Chernobyl; discharges from nuclear power stations and fuel reprocessing; and small amounts from consumer goods such as smoke detectors (americium-241), luminous paints, old gas mantles and some building materials. Occupational exposure of miners,

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radiographers and nuclear workers is monitored by law.

The total artificial contribution is small compared with the natural background, and the average worldwide dose is about 2–3 mSv per year, but because any avoidable dose carries a small risk, medical exposures are kept as low as reasonably possible and industrial sources are strictly regulated.

21.5 Explain how radiation exposure can be harmful and describe three key safety measures used to minimize its risks.

How radiation harms. Ionizing radiation carries enough energy to knock electrons out of atoms as it passes through matter, producing ions and free radicals. In a living cell this can break the chemical bonds of DNA: the cell may die (burns, or radiation sickness if enough cells are affected), or its code may be altered so that it divides out of control and becomes a cancer years later. Damage to reproductive cells can cause inherited defects. The bone marrow, gut lining and an unborn child are the most sensitive, the risk grows with the dose, and alpha emitters are most dangerous when breathed in or swallowed.

1. Shielding — sources are kept in lead-lined containers and handled behind lead, concrete or water screens, and workers wear lead aprons, because dense, high-atomic-number materials stop gamma and beta radiation. X-ray room walls are lined with lead.

2. Distance — the dose falls as the inverse square of the distance, so doubling the distance quarters the dose. Sources are handled with long tongs or in a remote cell and stored locked and labelled, away from people.

3. Time and monitoring — the dose is the rate multiplied by the time, so exposure time is kept as short as possible. Film badges and pocket dosimeters record each worker's accumulated dose against the legal limit, with warning signs, fume cupboards and strict hygiene — the ALARA principle, keeping exposure As Low As Reasonably Achievable.

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Points to Remember

- The Sun: 71 % hydrogen, 27 % helium; energy from fusion, core ≈ 15 million K.
- Planet temperature: intensity falls as $1/r^2$, but atmosphere also matters.
- Cyclones: convection over warm sea + Coriolis effect.
- Background radiation ≈ 2–3 mSv per year; radon is the largest natural source.
- Radiation safety: shielding, distance, time (ALARA).